Eurofound

Ethical implications of digitisation and automation in the workplace

Case study: Södertälje municipality (Sweden)

[Report that the paper supports]

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Case summary

This study investigates the application of two separate technologies within the municipality of Södertälje. The municipality has an overarching digitalisation strategy put in place in 2019 and municipal digitalisation unit acing as a driver for digitalisation and automation applications. In the strategy, the digital competency of staff is highlighted. Digital tools are also seen as a way of increasing work satisfaction, when used appropriately.

One application covered in this case study concerns the payroll office operating under HR and tasked with the salary administration of municipal employees. Here an RPA (Robotic Process Automation) technology is used for a multitude of processes, such as filling in Excel sheets regarding reported sick leave, habilitation endowments, termination of hourly contracts and filling in certificates. It was first introduced in 2018 for absence reports and later its use extended to further five processes: habilitation endowments, creating employer certificates, creating user accounts and clearing Excel-sheets of non-utilised 0-hour contracts

The other technology application is related to an AI chatbot used in the contact centre responsible for the customer service towards citizens. The unit operates under the communications office at the municipality. This unit introduced the AI chatbot on the municipality's website in 2019, where it answers questions posed to it in the chat. The AI is based on natural language processing, where it learns to understand questions and different formulations better the more it is used. The answers to the questions are formulated by employees at the contact centre. Since 2020, the AI chatbot is also connected to IoT sensors in the municipality and is thus able to inform about real-time water temperatures and parking space availability.

Measures put in place to cope with the ethical and working condition implications of the technologies included requirements posed for the technology developers regarding data protection and data minimisation, which is of relevance both because the data processed at the payroll office is employee data, but also since as public sector actors, compliance with GDPR is a central concern for the employees. Both technologies were built in collaboration with the employees at the units and only rolled out large scale when deemed to fulfil the function they were purchased for. Understanding how these processes work and how they process data increases the comfort in using them among the employees. The digitalisation strategy's overarching goal to provide staff with relevant competency of the technologies has been fulfilled in these two units. Overall, both the management and the interviewed employees describe working with these technologies as rewarding. Most importantly, employees have been involved from the very initiation to building the processes in collaboration with the respective technology developers with important implications for employee agency and human oversight of the technologies.

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Introduction

Background and objectives

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Methodological approach

This case study draws primarily from qualitative interviews conducted in the payroll office and contact centre at Södertälje municipality (Sweden).

Table 1. Profiles interviewed in establishment

Interviewee type	Department and fieldwork date
HR manager	Payroll office 09/03/2022
Line manager	Contact centre 10/03/2022
Employee representation	Trade union Vision, local union faction 26/04/2022
Innovation manager or technology developer	Digitalisation unit 21/02/2022 27/04/2022
Worker(s)	Payroll office 07/03/2022
	Contact center Former OHS representative 10/03/2022

Source: authors' own elaboration.

In addition to the qualitative interviews, the case study also draws from desk research and documentation provided by the management, supplementing the information about the establishment profile, the measure(s) taken to deal with the ethical implications of the technologies in focus and the impact on work organisation and working conditions.

Rationale for case selection

The case municipality is of interest as it is using digitisation and automation technologies (with AI embedded capabilities) in a multitude of functions and variety of applications following an overarching strategy for digitalisation of municipal functions. Machine learning is used in both external communication and internal processes within the municipality, whereby the employees are both affected in their work tasks as well as being themselves to a degree subject to RPA. The municipality has further extended its digitalisation to an IoT network with AI-sensors used in distinguishing bike traffic and use of garbage disposal stations. Sensors also measure temperatures on bathing sites and air pollution, feeding into

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datasets of open data, which are utilised by the chatbot covered by this case study. The case is also of particular interest as it affects services that are public in nature.

Report structure

The report is structured as follows. We will start by introducing the geographical context and give an overview of the two units in focus. We will then introduce the technologies and their implementation in the units in focus, as well the general digitalisation efforts made in the municipality as well as their positive and negative implications for working conditions. The ethical concerns raised by the interviewed staff is discussed in this context. The third section of this report introduces relevant sections of the digitalisation strategy driving the ethical implementation of these technologies as well as the competency heightening work that has taken place in the two units and the municipality as a whole. The report is finished with some final takeaways and lessons learnt as well as discusses future digitalisation efforts at the municipality.

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1. Establishment profile

Type of entity, geographic location and activities

Södertälje municipality is a local authority in Stockholm county, the capital region of Sweden. The city of Södertälje is first mentioned already in the 14th Century but formal structures were established in the 1860's with the Swedish municipal reforms. The current legal framework governing municipalities is the Local Government Act [Kommunallagen] 2017:725. Duties of the municipalities include basic schooling, childcare amenities, care services and recreational and cultural activities. Södertälje municipality also owns the company Telge with 14 daughter companies that provide municipal services such as housing, electricity and waste management. In 2022, the Södertälje municipality has 97,381 residents.

Size and workforce composition

The municipality as a whole employs almost 6,500 permanent employees and has another 2,000 employees on 0-hour contracts. The two units selected for this case study are the contact center and the payroll office

The contact center has a workforce of 21 persons, the majority which are women (16 employees). Five employees have temporary contracts and 11 have permanent contracts. The age span is of 20-63 years of age, while most employees (17) are between the ages of 20-39, 30-39 being the largest age group (10 employees). The contact center is tasked with customer service and advisory tasks towards citizens.

The payroll office has 18 employees, all women and all holding permanent contracts. 8 employees are between the ages 20-49, and 10 employees over 50-years of age. The office is responsible for administering salaries and endowments to municipal employees.

Organisational structure

Both offices are part of municipal administration.

The contact center is located under the communications department.

The payroll office is located under the HR department.

The digitalization unit is a small unit of one manager and three employees that has been created to follow up on the implementation of the digitalisation strategy.

Form of employee representation

The municipality has a collective agreement and a local union faction (in Swedish 'fackklubb') at the municipality. Local union faction is a committee of local trade union representatives affiliated to a specific trade union. The collective agreement regulates issues such as minimum wages and other renumerations, sick pay, working time arrangements, paid leave, terms for lay-offs and insurance. There are also occupational safety representatives. They represent employees in matters relating to occupational safety and health as well as work environment. Currently, neither the contact centre nor the payroll office has their own internal OSH-representative. The payroll office has OSH-representation in the general HR department, and the contact centre in the communications department. However, at the time of implementation of the chatbot and for a while after the contact centre had its own representative, who is still an employee at contact centre.

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2. Technology adoption and implementation

Type of technology and rationale behind adoption

Södertälje municipality has strategically developed digitalisation in municipal functions based on its digitalisation strategy. The digitalisation strategy started as a political decision in the municipal government in 2018 to develop the digitalisation and accessibility of municipal services. The strategy mandates each unit to integrate a unit specific digitalisation plan into their operational plans**.** The strategy states that follow-ups with units take place annually to ensure that all municipal functions have harmonised process in their digitalisation functions. The digitalisation unit arranges a monthly digitalisation forum, where representatives form all municipal units gather for an hour to discuss and inform others about their digitalisation efforts and find synergies between initiatives. There are also external experts invited to present and increase knowledge about digital tools such as artificial intelligence, although the purpose of such presentations is primarily promotional.

The payroll office uses a robotic application called Ragnhild, which performs tasks such as removal of non-utilised 0-hour contracts from Excel-sheets, absence reporting, and fills in certificates. Ragnhild is based on RPA technology, which is a 'narrow AI' application executing tasks based on pre-programmed rules rather than having 'intelligence' on its own or a capacity to learn. It can independently perform tasks that are conducted the same way each time.

The technology is a tool used in the everyday functions by the employees at the payroll office and where it assists them to process data regarding the general 6,500 municipal employees. The purpose of the RPA is to automatise repetitive tasks at the payroll office. For instance, the absence report for hourly employees (employees holding 0-hour contracts that are paid on the basis on hours worked rather than monthly salaries) used to be sent to an employee at the payroll office who then manually filled in the information from the Excel-file to their payroll system. Now, the RPA technology conducts the task of transferring the information from the Excel-file to the payroll system. This also reduces the chance of human error.

The contact centre of the municipality uses another AI-based technology, that is, the chatbot Kim, which serves as a 'digital coworker' that answers pre-determined questions from municipal citizens on the municipal website 24 hours a day. In the digitalisation strategy at the municipality, the utilisation of the chatbot is seen as a measure to increase accessibility for municipal citizens in a way that better corresponds to their daily routines. . This technology uses natural language processing, which means that the chatbot learns language, expressions and meaning and improves the more it is used. Kim got later connected to an IoT network around the municipality and provides citizens with real time data of water temperature on the bathing sites and free parking spaces in municipal parking lots.

The rationale behind the adoption of the technologies was to make service more efficient to the municipal citizens and ensure quality of service. For the payroll office and the RPA technology, automating repetitive tasks and saving employee time were important drivers.

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Based on interviews conducted with the digitalisation experts at the municipality one important benefit with digitalisation processes is that the work content of municipal employees better corresponds with the employee's education, and that they are not spending time conducting menial tasks such as administrative data entry in Excel sheets. Otherwise, they argued, the municipality might struggle to hold onto its qualified staff. Competency is one of the overarching focuses for development listed in the municipality's digitalisation strategy. While the strategy aims at removing repetitive administrative tasks and free time for employees to do tasks that correspond with their education, it is also recognised that both employees and managerial staff will have needs in regard to competency development and reskilling to cope with the technological change and the new role it poses to the units.

Approach to technology development and adoption

Both technology software are bought from an external service provider and then tailored to the needs of the unit. The applications of the RPA technology and the chatbot were then designed in collaboration with employees at the respective units. The municipality has its own digitalisation unit, which has strategic oversight over the digitalisation processes at all municipal units. They also arrange the digitalisation forum (discussed under measures) where all units in the municipality meet and share experiences.

Payroll office

At the payroll office, the adoption of the RPA technology was driven by the management after being introduced to the technology at a fair. The primary motivation for adopting the technology was its potential to eliminate repetitive and routine tasks, thereby improving the efficiency of payroll processes. By reducing the time employees spent on manual activities, such as transferring information from spreadsheets to the payroll system, the technology also minimised the risk of human error. The management at the payroll office contacted the digitalisation unit at the municipality to see whether RPA technology could be implemented at the unit. The payroll office was the first unit to implement digitalisation and automation processes in the municipality.

Only after they had been given an approval to procure the technology were employees at the unit involved in the implementation. It can thus be said that the decision to pilot the RPA process was management driven. However, employees at the unit were involved in designing the RPA implementation in collaboration with the technology developer as well as mapping and deciding on the processes that could be automatised in the unit. Although no local OSH-representation was present at the unit, one of the employees was appointed to be responsible for the technology deployment at the unit. They had made study visits to other municipalities that already utilised RPA technology to gain an understanding of what it does and how it works and could spread the knowledge to their coworkers. The responsible employee was tasked with mapping needs and feedback from her colleagues at the payroll office regarding the technology implementation and tailoring it to their needs.

They had detailed knowledge of the work processes targeted for automation and were responsible for identifying perspectives and concerns raised by their colleagues, as well as analysing potential challenges associated with the implementation of RPA technology. In consultation with managerial staff, a single process was selected as a pilot project. The selection criterion was that the unit wished to begin with a process considered to be 'sufficiently challenging', namely the processing of absence reports submitted by hourly

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employees through an Excel spreadsheet. After this initial implementation, the RPA technology has later been expanded to be used in both less and more challenging implementations. Thus, the implementation has been gradual.

According to the interview with the employee who was responsible for designing the RPA process utilised in her unit together with her colleagues, no direct concerns were raised regarding the deployment of the technology. However, the initial hesitation regarding loss of jobs had to be overcome and was a topic of dialogue. The employees had concerns regarding the automation of the unit as a whole and that their jobs would be made redundant. Reassurances from the management were made that it is not their ambition to lay-off staff after to the implementation of the RPA. Rather, the RPA is meant to ensure focus on more meaningful tasks. Some implication came up in the interview with the management regarding re-recruitment not being necessary due to automated processes after some of the staff have retired. Thus, despite the current staff having received reassurance regarding their jobs being secured, this does not necessarily mean that human labour at the unit will not be reduced in the future. Simultaneously however, the employee interviewed did not see that the RPA technology could replace a human worker and that their competence is still very much required.

The RPA technology was first introduced in the unit in 2018 in the absence reporting process and has since been expanded to the other four processes: habilitation endowments, employer certificates, clearing Excel sheets of non-utilised 0-hour contracts as well as the ordering of user accounts.

Contact centre

The process to adopt the chatbot technology was first initiated by the manager at the contact centre. It was discussed internally at the unit at unit meetings with employees and managerial staff. The staff also participated in information meetings with the technology developer to gain an understanding of what AI is, and what the chatbot can and cannot do and how it is operated. After being discussed internally, the unit applied for funding from the municipal administration for the technology adoption.

The unit developed and piloted the chatbot in close collaboration with the external technology developer. The employees at the contact centre made a list of questions they often receive from the public, and then formulated answers to the questions. In the beginning, the technology developer was going through the chatlogs and altered the questions and answers as per request from the employees from the contact centre. Today, the employees have access to a revision tool and have control over the process themselves.

During the pilot period the chatbot was only deployed on some sections of the municipality's website and extended to the whole website in 2019 when the employees were happy with the technology and deemed that it could answer questions accurately. The municipality's digitalisation unit was not involved from the outset of the chatbot deployment, but later contacted the contact centre to ask them if they were interested to integrate IoT into the chatbot functions. The chatbot got connected to the IoT network in 2020-2021 and can now inform on real-time data regarding water temperature on bathing sites and free parking spots.

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General views

According to representatives of the digitalisation unit interviewed for this study, internal interest in robotisation and automation technologies within the two units examined was essential to the successful adoption of these digital tools. One of the digitalisation experts interviewed described that the digitalisation unit cannot and should not force adoption of technologies in any unit. The staff at the units are the end-users of the technologies and have to since have an internal interest towards using tools such as RPA and AI: 'Thus, the initial spark for digitalisation must be anchored in the unit itself. For the payroll office, management had explained and discussed the RPA application with the employees. According to the union representative interviewed, the framing of the technology as a tool that can assist in tasks that are *'soul-destroying and boring'* was important to create acceptance among the staff. At the contact centre, the decision to adopt the chatbot was done in dialogue between employees and managerial staff.

Embeddedness of the technology in work processes and activities

Payroll office

The RPA technology at the payroll office was the first automation pilot at the municipality and its use has been expanded both to other units and to multiple tasks within the payroll office itself. It was first utilised in filling in Excel sheets about hourly worker's leave of sickness and leave of care for a sick minor. This task used to take up three working days each month when done manually, where it has now been reduced to one. One of the interviewees at the digitalisation unit also saw that utilising RPA results in quality improvements, as humans are more prone to mistakes in repetitive administrative tasks. This was however not highlighted in the interviews conducted with managerial staff or employee at the payroll office.

Today, the RPA at the payroll office has automated five processes. It is used to fill in Excel sheets regarding leave of absence among hourly workers and habilitation endowments (endowments for persons with functional disability participating in labour market programmes in accordance with the law on support and service for certain disabled (1993:387) [Lagen om stöd och service för vissa funktionshindrade]), automatically terminates hourly worker's contracts after a period of not taking shifts, prepares employer certificates [arbetsgivarintyg], and prepares user accounts for external consultants conducting tasks commissioned by the municipality. The satisfaction with the technology was high among managerial staff and employees at the payroll office alike as it has taken over repetitive tasks that take up a lot of working time. Thus, the use of the technology has been expanded to another four processes gradually.

The RPA now handles Excel files regarding habilitation endowments for employees. The task conducted is technically simpler than the first RPA implementation, and thus the second RPA application at the unit was adopted and run without problem. Conversely, both the employee and managerial staff at the payroll office have their hesitations regarding the deployment of the RPA technology to filling in employer certificates [arbetsgivarintyg] for employees through their webservice. The employer certificate [arbetsgivarintyg] is a particular certificate used in many functions in Sweden. It is essentially a certificate of employment, showing where the employee has worked, for how long, and how much they have earned.

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The filling in of employer certificates task is a lot more advanced for a robot to do than the preceiding RPA applications at the payroll office. It has thus required a lot more oversight than the other processes where employees have to assist the robot in filling in the certificates as well as quality control. The employees also have to confirm the certificate that the robot has filled in using the BankID system [a Swedish system for online secure identification], which is not something that can legally be given to a robot.

The RPA is also used to automatically clear employee data when contracts of hourly workers (i.e. employees at the municipality that are employed with 0-hour contracts) when they have not taken shifts in six months by removing their data from an Excel-list. Interestingly, this process is now completely externalised to the technology developer, who asked the payroll office for support in building their service for data clearing from Excel sheets that they can sell to other clients. The payroll office purchases the whole service of clearing the Excel sheet regularly directly from the developer, so that they receive the end product whereas the other four RPA processes are run by the payroll office themselves with assistance from their internal tech support and the municipality's digitalisation unit.

The RPA process of clearing unused 0-hour contracts was externalised as this saved time and human resources both as the employees at the payroll office do not have to do updates to the RPA system themselves. This entails that every time there is a software update to the RPA system, the payroll office does not have to oversee their RPA application in relation to the update but this work has been externalised to the technology provider. The union representative interviewed saw that the benefit of using an external provider for such a function is that employees are being relieved of the task of running the process and monitoring the robot. For the payroll office, this means time-savings as they do not have to deal with updates as they do with their other four processes. Updates can also entail that the robot needs to be reprogrammed if there are any changes in the internal payroll office systems, as the robot then no longer knows where to transfer the data form Excel sheets.The manager saw it as possible to externalise additional processes if the technology developers had an interest in doing so.

Contact centre

The chatbot Kim has been deployed on the whole webpage of the municipality since August 2019. It was first piloted on the Culture and Leisure Office's part of the webpage, then gradually rolled out to the Building Permits Unit part and to some part of the Education unit's part of the website. The gradual rollout was done to ensure that the chatbot answers in a way that is satisfactory to the contact centre. Questions, answers and tonality of chatbot language were all fine-tuned at this stage.

Since 2020, the chatbot is connected to sensors that can provide real-time data regarding water temperatures on bathing sites as well as available parkin slots in the municipality. The employees are themselves in charge of formulating the answers to expected questions as well as supporting other municipal offices in utilising the chatbot to answer questions related to their services. What the chatbot's artificial intelligence is used for is to understand the questions correctly, learning synonyms and different ways of expression as well as learning to understand content despite of typos. Out of the employees at the contact centre, four persons are actively working with the chatbot – going through logs and formulating and reformulating answers the chatbot gives as well as communicating about the chatbot to other municipal

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offices. One person acts as a team lead and thus is the contact towards the technology developer. In addition, today two out of the four persons working with the chatbot at the contact centre are in charge of training both internally to new colleagues and giving introductions at other units about AI, the chatbot and how it is used. This is so that other municipal units are able to utilise the tool in thinking about questions that the chatbot could answer and updating the contact centre if there are any changes in their services so that the contact centre can ensure that the chatbot's information is up to date. The contact centre's number of staff has grown since the implementation of the chatbot and working with developing the chatbot has been priority by management. The employees working with the chatbot feel like they have been given time to conduct the administration and development of the chatbot. The growth of the unit cannot be assigned only to the chatbot however, since the unit has also extended to other new functions simultaneously. Nobody is working only with the chatbot but are also in direct contact with municipal citizens.

Impact of the technology on working conditions

Payroll office

The RPA technology at the payroll office is characterised by all informants as something that saves employee time and assists in peak workloads which take place during the time of the month when salaries are paid out. The timesaving offered by the different processes vary however between the processes. Especially the process where the RPA fills in employee certificates which runs on daily basis has shown itself to not really be timesaving, as the RPA process requires a lot of assistance from the employees. Thus, the process has relieved them of this task although some of the manual work is now automated.

At the payroll office it is perhaps more apparent that job content has changed to more the job profile of a controller than an administrator in that there is less manual work and more coordination, overseeing and management tasks. For the employee responsible for the building and tailoring of the RPA application at the unit, work content has changed as they are now tasked with the maintenance of the RPA system and any required updates. This was seen as a downside of the technology implementation, as system updates would require continuous dialogue with the digitalisation unit at the municipality. This has thus resulted in an increased dependency of the technological knowledge located outside of the unit, which in turn has driven the process of externalisation further as one of the processes is now ran completely by the technology developer.

Satisfaction with the technology as a tool that saves time and frees worktime for other, less repetitive tasks is high as long as it works as it should. Work burdens may increase in cases where the RPA technology has stopped functioning according to plan and the unit has had to return to conducting the filling in of Excel sheets manually. This has of course caused a problem at the time, especially as worktime has not been scheduled for having to conduct the tasks manually in case of system malfunctions. RPA processes that are connected to salaries are only run during the time of the month when salaries are being paid out, i.e. the busiest week of the month for the employees. When these RPA processes are not functioning as expected and the system crashes, the workload does fall upon the payroll office. Paying out salaries to municipal employees has to happen in a timely manner, and thus cannot be delayed until the RPA process has been rebooted and the problem causing it resolved. When this

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happens, employees at the payroll office have to go back to entering data manually which can be stress-inducing.

These kind of problems with bugs and malfunctions in the RPA systems are especially connected to the beginning of the implementation period and solved on an ad-hoc basis. Once solved and the RPA systems refined to be more reliable and robust, the most stressful time of the month (when salaries are to be paid out) has a little lighter work burden due to the RPA. The problem with the introduction of technology can thus be summarised in that it is designed to work, and employee worktime has been scheduled accordingly to the utilisation of time sparing technology. Some problems do however only reveal themselves after implementation and thus the workload falls back on the employees. The general assessment from both the managerial and employee perspective is that once such initial problems have been solved, the technology is time sparing and relieves employees of data processing tasks.

Contact centre

Based on the interviews with the line manager and the former OSH representative at the contact centre, the chatbot has been well received at the contact centre and perceived as a rewarding tool to work with. The amount of contact with the public in terms of phone calls has not changed, but based on statistics that the contact centre collects regarding chatbot use, many citizens seem to use the chat after the opening hours of the contact centre, i.e. evenings and weekends. According to the trade union representative, there is likely some user groups that are more comfortable with utilising the chatbot, which allows for more time to be allocated to those that call in, such as elderly persons. One of the digitalisation experts interviewed analysed that as the chatbot knows how to answer simple and straight-forward questions, such as questions regarding museum opening hours in the municipality, the persons working in the contact centre will now primarily receive the questions that are more complex and interesting. However, the persons at the contact centre say that counterfactuals are difficult to establish. It is hard to know how many calls they would receive without the chatbot being in place. They have however not experienced that some types of questions would have disappeared from the phone lines after the chatbot was put in place.

The employees do, however, view the chatbot as a valuable support tool during periods of crisis, such as the COVID-19 pandemic, when it can rapidly direct citizens to the appropriate external webpages through embedded links. Furthermore, following its integration with an IoT network, the chatbot has been able to provide services that would be difficult for human employees to deliver efficiently, including real-time information on water temperatures at bathing sites, air quality levels, and the availability of parking spaces in parking facilities.

For the employees, the deployment of the technology has increased their digital competence in regard to what RPA and AI can and cannot do as well as how such processes are built and maintained. The increased digital competence that they have gained through the deployment of the chatbot is something that can be to their benefit on the labour market and thus increase their employability. As the team leader who originally started working with the chatbot at contact centre described: '*I am a social scientist and not particularly tech-savvy. It has been fun to learn what AI is, how it works and how the digital development can contribute to the citizenship'.* Generally, the line manager interviewed described the role of the contact center to be increasing accessibility to information and municipal services for citizens, and thus digital tools can be used for this purpose.

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The deployment of the technology has also contributed to changing the organisational structure as a new role as a development manager [verksamhetsutvecklare] has been created where there used to be no middle-management or other roles between the manager of the unit and the employees. This has allowed for more room for career development internally at the unit.

Ethical concerns about technology use

Payroll office

Among the employees at the payroll office, some concerns regarding loss of jobs were originally raised. The management made promises about job reductions not taking place and they have not done so since the initiation and spread of the RPA technology since 2018. It is however less straightforward whether re-recruitment will be needed after some of the current employees retire due to processes being automated. The RPA process gives some effectivity gains, but the sense among all informants interviewed for this study is that human intelligence and employee competence will not be replaced by the technology. The digitalisation expert interviewed described the RPA as a 'stupid' AI, which can only learn tasks that are conducted the same way every time. As the employees at the payroll office have themselves mapped the processes where they see potential for automation, the selection has focused on tasks that the employees have found tedious (which are also on the level that an RPA can learn to do). The trade union representative described the process of clearing Excel sheets of unused 0-hour contracts was a task that got continuously down-prioritised. The employee interviewed at the payroll office also describe that those co-workers that were more reluctant towards the RPA technology initially became accepting of it when seeing what it can (and cannot) do, i.e. as a tool that facilitates their work rather than replaces it and threatens their competence and livelihood.

Based on the interviews, neither the manager, the trade union representative nor the employee from the payroll office have heard any concerns raised from the general 6,500 municipal staff. As it is still a human that has control over the data processed and the final decisions made by the RPA technology, the employees whose certificates are being pre-filled in and whose salaries are being processed with the RPA technology remain unaffected. It is no secret that RPA is in use – the functions in which it is used are displayed publicly on the municipal website and it was internally communicated to the employees at the municipality that the payroll office is starting to use RPA as a tool. According to the trade union representative, the human oversight and human-in-the-loop principle were important to communicate about. They had heard from other municipalities, that had started investigations into how algorithmic decision-making is being used. They analysed that this has not been an issue that has required attention in Sodertalje municipality, as the robot only processes information and creates documents, but the final decisions are always made by a human employee.

The municipality is loud of their digitalisation processes, and the general approach is that municipal units are encouraged to implement technologies. As such, the decision regarding the implementation of RPA can be taken by the payroll unit itself with support of the municipal board and the digitalisation unit without any large-scale consultation of the municipal staff-force. The monthly digitalisation forum arranged for all units is also a channel

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to communicate, discuss and inform internally about digitalistaion processes taking place in the municipality.

The human control over the RPA processes and the utilisation of the technology in tasks that there has always been reluctance towards has positive implications for the acceptance of the technology. It is viewed as a tool that does not compete with human intelligence. As it has also proved itself to relieve employees of this task in the simple processes were it functions well. Both the employee at the payroll office interviewed and the manager did however recognise that not all employees were overjoyed by the change and there are those that would have preferred to not have their job profiles changed from administrative tasks to monitoring the robot.

The final responsibility for any mistakes that the robot might make lies with the manager, and finally the HR director. This has however not been something that the manager has felt worried about, due to the sense of accountability and human control over the RPA processes that they have gained from building them themselves at the unit and approving them only when they felt that they were fulfilling the function based on the needs mapped at the unit. This despite the fact that there has been hick-ups after implementation which has momentarily resulted in workload peaks and that there is now some reliance on functioning technological support in routine tasks.

In essence, understanding the technology, what it does and does not do and why, appears to reduce alienation from it.

Contact centre

At contact centre, central concerns have been GDPR and data protection as well as tonality of the chatbot. Data protection *'was and is very topical'*. There were concerns about data being stored externally at the developer and the contact centre being held responsible for any breaches. According to the line manager interviewed, this was of particular concern due to their role as a public sector actor. The staff at the contact centre discussed extensively where the data is stored, what kind of sensitive data might come up in a chat and how security can be ensured. Measures for data protection and data minimisation were implemented, described in more detail under the measures section.

The concern among the employees among the contact centre regarding the use of the chatbot was whether the AI could get the tonality right. Citizens reach out to the contact centre sometimes in matters that are rather serious in nature. Sometimes the chatbot had only recognised one word in a quite serious context, and answered in a way that was tone-deaf. The employees at contact centre have quickly responded asking the technology developer to correct this. Simultaneously, they wanted the AI to answer in a humanlike way. Finding a balance between a flowing dialogue and the AI's tone not being overly light and humorous was a much-discussed issue.

'*Not every person is super digital. Some may find it unpleasant to talk to a robot, if it's very mechanical and strange. We want it to be a good dialogue. Absolutely not that the person should think that they are talking to a human, that is not what I mean. But some might not like talking to a machine.'*

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Employees at the contact centre explained that, as a public sector organisation with a responsibility towards citizens, it is important that they are "on top of things" and can trust the AI system in its communication with the public. These concerns have been mitigated through the involvement of the municipal lawyer in ensuring that GDPR requirements are built into the system, as well as through employees' ability to oversee and influence the chatbot's responses and tone of communication. Here, too, employees retain autonomy and agency over the AI system, which is viewed as a complementary tool that supports their pre-existing tasks rather than replacing them.

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3. Measure(s) addressing ethical implications of technology

Type of measure(s) and design

There is an overarching **digitalisation strategy** at the municipality. The strategy states that digitalisation processes need to be scalable, developable, and manageable. Scalable means that the digitalisation measures should be able to increase in volume and reach more people and realise more benefits. Developable in the sense that they can be adjusted if the organisation and organisational functions change and the organisational needs change. Manageable entails that they should be able to be sustained by the municipality without additional costly or complicated inputs. This is to ensure that the digitalisation processes are flexible to correspond to changing needs as well as not requiring complex maintenance. One of the key components of the digitalisation strategy is increasing the digital competency among the employees and managerial staff. The strategy envisions that municipal staff will be relieved from repetitive tasks and to a higher degree conduct work that corresponds with their education. It is further stated that training to keep up with robotisation will be offered where needs occur. One such way is to utilise the digitalisation forum, but the digitalisation unit is primarily tasked with overarching competency development. On top of this, each unit has their own initiatives, but the implementation is uneven.

The digitalisation strategy is connected to an overarching goal at the municipality to be an attractive employer. According to one of the digitalisation experts interviewed, those that work in the payroll office and most of those working in the contact centre have a university degree. Thus, if their work content is used to answer the same question regarding opening hours of a museum or manually transferring data from Excel sheets, it is seen as a considerable risk that they will find other, more meaningful employment. On top of development of competence to work with and lead development processes, the municipality sees digitalisation processes as a way to offer employees appropriate digital tools that contribute to an attractive work environment where they can use their working time in a meaningful way.

On top of this, there are **regular meetings for knowledge-sharing and problem-solving regarding the technologies between the municipality's digitalisation unit and all the units that have adopted some automation technologies such as the RPA or the chatbot.** The units have a forum where they can discuss and follow-up how the technological implementations have worked and whether they have encountered any problems. If there are problems or improvement suggestions, the digitalisation unit contacts the technology developer. According to the employee at the payroll office, they have even found these meetings more frequent than has been necessary, as they have been quite involved in the planning of the RPA processes from the outset and were thus familiar with the technology and did not feel like they gained much from the meetings. Further, as the different units utilise very different technological applications, knowledge sharing is not always meaningful. The meetings were monthly in the outset of the technology implementation but have now been conducted on a bi-monthly basis.

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Both units in the case study **posed requirements on the external technology developers.** At the payroll office, a protocol for data protection was created with the technology developer. Further, as the RPA technology is handling contractual data, the rules in employee contracts and thus collective agreements have had to be in-built into its' processes regarding premises on which sick pay is paid out, for instance

Among the employees working with the chatbots, central concerns have been GDPR and data protection as well as tonality of the chatbot. The municipal lawyer was involved in the implementation process to ensure that the implementation and use of the chatbot happens in accordance with Swedish law and regulations. The union faction was also consulted in the implementation, which is described as standard procedure at the municipality in change processes. This consultation processes and dialogue with the technology developer increased the trust towards the digital tool among the employees that the chatbot would not create problems for them in the form of discontent among citizens.

In the interview with the former OSH representative at contact centre it was described how the utilisation of automation technologies **is incorporated in the systematic OSH-work** and monitoring at the unit, including a regular work environment survey with incorporated risk analysis. The OSH-work is however holistic rather than technology specific, including both physical and psychosocial factors related to the work environment in accordance with Swedish regulations for systematic OSH-work[1]. However, the OSH-representation has a role in considering the work environment implications of any new changes, for instance what an added system might entail for employee welfare. During their time as an OSH-representative, no negative effects of the chatbot came up in the surveys nor had any complaints been filed to the OSH-representative.

Training on an annual basis is provided by the external service provider are offered for employees that work with the chatbot regarding AI processes: what AI is and how the chatbot functions. There are annual webinars where participants from across the Nordic municipalities utilising the chatbot technology gather for gaining more competence and knowledge-sharing. Each time a new municipality deploys the technology, the technology developer invites representatives from all previous municipalities to the webinar. According to the line manager, it is possible for an employee to participate multiple times in the formal training offered by the technology developer if an update is wanted. The trainings focus on increasing the understanding of the AI and the chatbot, explain how the chatbot used in the municipality differs from other chatbots, what the technological solution for this particular chatbot is and how the tool for formulating and reformulating answers works. These trainings are of significance to increase human agency over the technology, i.e. that the employees understand how the technology is making the decisions as well as how and why the technology is deployed. The training is offered to those employees that operate the chatbot.

Internally, employees working with the chatbot's logs and revisions has increased to from one to four. Originally, these employees were trained by the technology developer. Now the contact centre has ownership of the training content for their staff, and training takes place peer-to-peer at the workplace. Further, the contact centre holds **presentations about the chatbot to other units,** mainly focusing on persons that have a developer or line manager

[1] Provisions for Systematic Work Environment Management [Systematiskt arbetsmiljöarbete] (AFS 2001:1),

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role. Here, the line manager interviewed describes the importance of being adherent toward the target group needs for training content and level. Some participants in the units where the contact centre have held presentations have found these type of technologies as difficult to use and even a little intimidating. Thus, content needs to be adjusted with the aim of increasing comfort level of those participating in the presentations.

At the payroll office, the processes were designed collaboratively by the technology developer and the office staff. The technology developer, the digitalisation unit, and payroll office employees held a series of workshops aimed at designing and refining the automated processes. These workshops also served to contextualise the adoption of RPA within the unit, helping employees understand the rationale behind its implementation. All staff members were included in these activities. In addition, employees compiled a list of processes they believed could be automated. Consequently, employees were actively involved in determining which tasks should be assigned to the robot.

Staff was trained by the technology developer in the use of RPA in the building process. During the time of process design, the technology developer assigned one of their staff to be physically located at the payroll office for support, assistance and being able to answer questions regarding the technology. The employee responsible for the RPA at the payroll office created a manual for her colleagues in how to use the RPA. Thus, the staff is knowledgeable in running the RPA processes and have a good understanding of how the RPA functions. However, technological bugs and crashes require external support from tech support at the municipality and the digitalisation unit.

Implementation of the measure(s)

The GDPR contract the contact centre negotiated with the technology developer agreed that all data deemed sensitive, such as names, personal numbers, addresses etcetera will be removed after an hour. In general, the questions will be stored for a month. After a month has passed, the only data stored are categories of questions for statistical purposes. A downside of in-built data protection and data minimisation is that human operators will not have a chance to get back in touch with the citizen's if they would like to add something or change the chatbots answer.

Regarding concerns about tonality of the chatbot, this was mitigated by the contact centre being themselves in charge of formulating the answers to the questions. What the chatbot's artificial intelligence is used for is to understand the questions correctly, learning synonyms and different ways of expression.

Not applicable since no monitoring mechanisms were introduced.

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4. Outcomes of the measure(s)

Impact on working conditions

Out of the measures discussed above, the most important for working conditions and ethical introduction of technologies are the overarching education, training and knowledge sharing initiatives that have been implemented in parallel with the technologies. The general impression is that the units have agency and autonomy regarding the technologies, that the implementation has corresponded to needs and interests in the units. It is possible for them to withdraw from the technology use if the technologies would not fulfil their promise. Further, ownership and autonomy are positively supported by the agency the units and their staff have over the processes. This is much due to employees being deeply involved in building the processes in collaboration with the technology provider and having an understanding of the functioning and purpose of the underlying technology.

In both units, the introduction of the technologies has seen overall as a primarily positive development that has increased the competences of staff regarding automation processes, what they can and cannot do and how to manage them. Especially the contact centre that use a technology that learns and improves with use, they need insight into which terms the chatbot might be misunderstanding and therefore giving answers that are strange so that they can amend the problem in a good way.

No particular salary increases, or bonus has been implemented with the increased competence and not something that the employees have asked for either. At the contact centre, both interviewees described working with the chatbot as something intriguing and fun as it is a new method of reaching out to the citizens in the municipality, while the employees have had the opportunity to learn about AI and how it functions. At the payroll office, the experience the employee had in building the RPA processes and the experience they gained was considered a reward in itself. .

Impact on fundamental rights

Due to the initiative and involvement of the employees, it appears that the automated processes have been designed with respect to human dignity. The persons interviewed from the units did not see that the value of their work was diminished, or their competence belittled by the introduction of automation processes. At the payroll office it is perhaps more apparent that task content has changed to more controller tasks from administrative tasks. Although the employee interviewed saw this as something positive, they did also recognise that this may not be the case for or their colleagues. They analyse that some may find that the controller tasks are tasks that should be conducted by the manager rather than the employees and thus do not belong to them.

As for the employees impacted by payroll office RPA, based on the interviews of payroll office staff there has not been much feedback – positive nor negative. It seems like most of the 6,500 employees at the municipality have not paid much attention to the fact that RPA is used, and in which functions it is used. This is seemingly due to the limited impact of the technology on the daily life of the employees. Further, according to the trade union representative interviewed, it was clarified in the internal information sent out to employees

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that the robot does not make decisions, i.e. that there is still human oversight by payroll office employees. What is significant is that the final responsibility lies with a human operator, which impacts the human control over the technology. This was highlighted by both the payroll office manager and the line manager at the contact centre – that they have a responsibility toward the technology end-users as they have made the decision to implement such. However, this responsibility does not seem to weigh heavily on the managements shoulder, perhaps primarily due to being knowledgeable of the process as it has been tailored together with the employees at the payroll office. The manager in the payroll office describes the process of building and tailoring the technology together with the technology developer as something that increases trust in the process and describes a sense of control as they know exactly what the robot does

Other ethical principles and values

Knowledge and understanding of the processes the RPA and AI technologies do is significant for the trust towards these automation systems.

In both the RPA at the payroll office and the chatbot at the contact centre, the principle of human in the loop is ensured. At the payroll office, the RPA technology creates documentation that is always checked and accepted by one of the employees. According to the trade union representative interviewed, it has been communicated clearly when announcements of the RPA use in payroll functions were sent out to municipal employees that the robot does not decide whether salaries are paid out or not, but only assists the human worker at the payroll office.

In the contact centre, the employees have access to a revision tool where they can revise the chatbots answers if they see from logs that it does not understand the question correctly. Thus, the human operators have control over the chatbots answers which increases the sense of agency over the technology.

The people from the digitalisation unit interviewed for this study described that municipalities in Sweden are much slower at using AI as a tool in operations due to concern of robots making decisions – a concern that they deem that could be mitigated with an increased understanding of the technical functions. The RPA only 'pushes buttons in the order we tell it to push those buttons'. The robots in the municipality do not (currently) make decisions but act according to certain rules programmed into them and need to have human oversight. They deliberately avoid applications that they consider might be found controversial. It is however notable that digitalisation efforts at the digitalisation unit are primarily driven by an effectivity narrative and that within this unit the approach to technology is rather uncritical. The digitalisation unit representatives described a certain frustration that the full promise of automation processes is stalled as it against Swedish law[2] to utilise automated decision-making in municipal functions. An example given was that with construction permits, technically it would be possible for an AI to automatically accept

[2] Note: This law has however recently been revised so that municipalities can utilise automated decision-making. The law will enter into force in July, 2022 (Government Offices of Sweden, 2022). It is expected to become a significant issue for the municipalities to solve within which functions it is ethical to use automated decision-making and how it should be governed in a transparent and accountable way.
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construction permit application concerning a zone where building permission would not require any further investigation. However, in the current law sending out an automated acceptance is considered to be a decision (although based on pre-determined rules set by a human) and thus it cannot be used in such functions.

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Lessons learnt and takeaways

Success factors

The success factor for both of these cases is that the employees at the payroll office and the contact centre have a good understanding and knowledge of the automation processes utilised in their unit. Both units in the municipality initiated the piloting and deployment of the automation technologies themselves. The manager at the payroll office contacted the digitalisation unit rather than vice versa. However, the employees were only informed after the decision regarding the first RPA process had been made. Familiarity

Conversely, in the contact centre the initiation of the process to apply for funding for the chatbot was based on dialogue between employees and managerial staff. At the payroll office too, employees have been consulted regarding the processes they can envision as automatised.

The deployment in both cases has also been done gradually and have thus been manageable, as the expansion of the technologies to new tasks at the payroll office, or the deployment of the chatbot on the whole municipal website have only been done after the employees have been content with the previous process.

Generally, the municipalities strategy for developing employee competence of the technologies appears to be functioning well as the staff at both the payroll office and the contact centre described that they felt in control over the technology as they have tailored it to their needs themselves. This notion of control is interesting, as the representatives of the digitalisation unit on the other hand argue that there is no capacity for having internal AI or RPA experts in the municipality and that is preferable to purchase the service from an external provider and hire consultants and thus be able to better respond to the 'moving target' technological progress provides.

Both units thus feel like they understand and have autonomy over the technology, as they have been tailoring the processes themselves while simultaneously been increasingly tied to external technological expertise in their work functions. A mitigating factor regarding this problem may be that neither the chatbot or the RPA technology are novel technologies anymore, which allows the units to change provider or resign the contract if they are no longer content. At the payroll office, the service provider got swapped when the employees at the unit did not receive the support they needed.

Barriers to implementation of measures

No significant barriers to the implementation of either the technologies or the measures were recognised in either case. As elaborated on in this case study, both the payroll office and the contact centre showed internal interest towards piloting and finally adopting automation technologies. It has become apparent in both the interview with the digitalisation unit and interviews with the contact centre that is now conducting trainings internally and presentations to other units regarding AI, that the importance for internal 'maturity' towards digitalisation cannot be understated. The units must themselves want to implement digital tools, and management have to have anchored and ensured acceptance among the staff in order to adoption of digital tools such as RPA and AI to be smooth. Similarly, the interviewees representing the digitalisation unit have underlined that implementing

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digitalisation processes cannot be externally driven by the digitalisation unit despite their mandate to have strategic oversight, drive and harmonise digitalisation processes in the different units in the municipality. Rather, the units need to have an internal interest and an acceptance among the employees towards piloting and adopting digital tools such as RPA and AI.

According to the interviewee from the contact centre, they get a lot of questions regarding the chatbot from other municipalities, but many feel hesitant to change and implementing novel technologies where the implementation may require a lot of work initially before the AI learns to answer questions properly. As we have seen in both of the case of the contact centre and the payroll office, refining of the technological applications has been needed before they start functioning as tools for the employees. At the payroll office, both the manager and the employee interviewed has their hesitations whether the RPA process that fills in employee certificates has actually been worth the investment. This process has proved itself to be so complicated for the RPA software to conduct, that the particular process does not really free up any working time or decrease the workload for employees in the task. Similarly, as the RPA processes have crashed during the worst work peak, the workload has fallen onto to the unit as no time had been scheduled to conduct the task manually.

For the contact centre and the AI utilising natural language processing this learning procedure is perhaps more justified and expected, as it is in the nature of the technology to improve the more it is used. Thus, on the flipside, the AI technology is in a sense a work in progress. Thus, the AI chatbot as a tool to assist the employees at contact centre in their work is developing all the time. The issue with tonality was something that caused the employees' concern. Initially this unit had to rely on the technology developer to fix identified issues with tonality that caused concern among the employees. Now the unit can change the answers the chatbot gives themselves, which allows for more autonomy over the process.

The refining period that follows the implementation of the technology makes initial implementation labour-intensive and occasionally frustrating. The interviewed union representative said that in some units within the municipality where there is a heavy workload of repetitive administrative tasks, the union has probed in the potential to implement RPA technology to life the burden. These overburdened units have however felt like to process takes up too many already strained resources. RPA technology is thus not always in the reach of units where it could bring the most benefits.

The union representative interviewed for this study reflected that the union should have taken a more proactive role in the municipality's digitalisation processes. To date, the union's involvement in digitalisation and automation initiatives has been limited, apart from a number of information meetings during which representatives from the digitalisation unit were invited to present the municipality's digitalisation efforts and the technologies being implemented, as well as to discuss their potential positive and negative implications for working conditions.

According to the union representative, the substantial changes to working conditions brought about by the COVID-19 pandemic demanded significant attention from both union representatives and occupational health and safety representatives. As a result, digitalisation processes and their implications for employees were not given sufficient focus.

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Future plans (in the area of ethics)

The municipality's digitalisation efforts continue to progress in line with the priorities set by the municipal leadership. The IoT network currently utilised in waste management is planned for further expansion, while the use of RPA technology at the payroll office is also being extended to additional processes.

The chatbot has become an integrated part of the contact centre's operations, with some employees now working primarily with the chatbot. Efforts are currently underway to improve the integration of information from different municipal units into the chatbot's database of questions and answers.

As digitalisation and automation continue to play an increasingly prominent role within the municipality, and as working life gradually returns to normal following the COVID-19 pandemic, the trade union Vision has begun developing a policy statement on digital working life. At the municipal level, one of the digitalisation experts interviewed for this study argued that there is room for improvement in the municipality's follow-up of technology implementations, particularly regarding their impact on employees and staff satisfaction. However, no concrete plans for how such follow-up activities should be conducted have been established at this stage.

The digitalisation unit has ambitious goals for the rapid and large-scale implementation of digital technologies. In addition to the public-facing chatbot currently in use, the unit would like to introduce an internal chatbot for municipal employees and expand the number of RPA-enabled processes across the organisation. However, it is noteworthy that the authority to decide whether such technologies are implemented remains with the individual units, many of which have declined these proposals.

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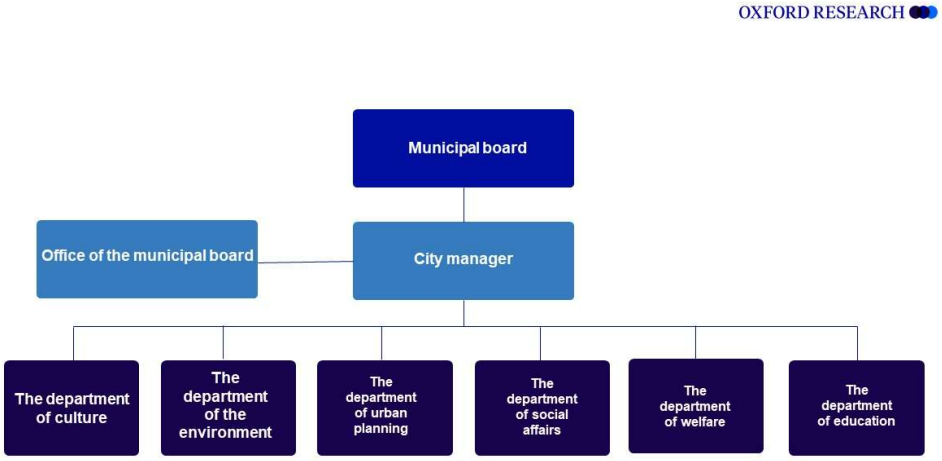
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Annex 2: Organisational chart of Södertälje municipality

Picture source: Södertälje municipality



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